

Pressure/Temp Cal Data Capture

Bench Tool User Guide

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KJO Liquid Rocket Motor Control System

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Standalone bench-test firmware, flashed onto EMU's physical hardware, that logs raw chamber/tank pressure and thermocouple readings to the SD card on a button press, for deriving pressure and temperature calibration curves.

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1 Purpose

This tool captures a single averaged reading of chamber pressure, tank pressure, and both thermocouples to a CSV-style file on the SD card every time Button C is pressed, timestamped from the real-time clock. It is meant for bench calibration work: apply a known reference pressure or temperature, press Button C to log the corresponding raw sensor reading, repeat at several reference points, then fit a calibration curve (slope and intercept) from the logged data offline.

It runs on the same physical EMU hardware as production firmware, but is a completely separate, standalone build – **no LoRa, no CAN**; it never communicates with RCU, GSEMU, or LCMU.

2 Hardware Required

- EMU hardware: Adafruit Feather M0, SH1107 OLED, MCP23X17 GPIO expander (Button C only), ADS1015 external ADC, two MAX31856 thermocouple amplifiers, PCF8523 RTC, SD card
- Both chamber and tank pressure transducers connected to the ADS1015 (channels 1 and 2 respectively)
- Both thermocouples connected (TC1 = chamber, TC2 = tank)

Note

This build shares EMU's physical SPI bus with the (unused, in this firmware) LoRa radio module. `RFM95_CS` is deliberately held HIGH before `SD.begin()` in `setup()` to avoid bus contention – no operator action is needed for this, it's automatic.

3 How to Use

1. Power on the board. The OLED shows boot status for each subsystem (OLED, thermocouples, GPIO expander, ADC, RTC, SD card) as it initializes.
2. On successful SD init, a fresh data file is created (`Cal_N.txt`, auto-incremented – see §4) and its name is shown on the OLED.
3. Apply a known reference pressure and/or temperature to the sensor(s) under test.
4. Press **Button C**. The tool averages 10 samples from each of the four channels and appends one row to the data file; the OLED shows the running sample count ("sample N").
5. Repeat steps 3–4 at as many reference points as needed for a good calibration curve.
6. Remove the SD card (or copy the file off) and fit the calibration curve offline.

4 Output File Format

Files are named `Cal_N.txt` (N auto-incremented from 0 each power-up, never overwriting an existing file) in the SD card root. Each file begins with two header lines, then one data row per Button C press:

```
09/09/2026
Time, Chamber P, Tank P, Temp 1, Temp 2
10:15:03.142, 812.40, 3021.75, 71.32, 68.90
```

- **Time** – HH:MM:SS.mmm from the RTC (date is recorded once, in the file’s first line, not repeated per row).
- **Chamber P / Tank P – raw ADS1015 counts** (10-sample average), *not* PSI. The factory PSI conversion is intentionally bypassed here – raw counts are what a calibration curve is fit against.
- **Temp 1 / Temp 2** – thermocouple readings in °F (10-sample average).

5 Notes

- If the RTC failed to initialize, the file’s date header reads “RTC not available” and each row’s time is a raw `millis()` count instead of wall-clock time.
- A failed SD write is reported on the OLED ("**write FAILED.**") rather than silently losing the sample – retake it if this happens.
- Status/boot messages are shown on the OLED only; they are *not* written to the data file (the file exists solely to hold calibration rows).

6 Version History

Version	Date	Notes
0.1	September 2026	Initial user guide.
